

1 Background

1.1 The double vision problem for Hintikka semantics

Hintikka semantics for 'believe':

- (1) $\llbracket \text{believes } P \rrbracket^c(w, y) = 1$ iff $\llbracket P \rrbracket^c(v) = 1$ for all $v \in \text{Dox}(y, w)$.

For short: to believe that P is to *doxize* the proposition that P .

M&S suggest that examples like the following are problematic for the Hintikka semantics:

- (2) Joe believes that Eve is involved with a linguist.

(2) can be judged true in a case where Joe is looking at Eve dancing and has the internal monologue 'I bet that dancing person is involved with a linguist, unlike Eve, who is not even at this party and not involved with anyone'.

M&S treat it as uncontroversial in this case that Joe's doxastic set contains worlds where Eve is not involved with a linguist, so that it's a counterexample to the Hintikka semantics. I don't think this should be uncontroversial: *doxastic set* is a theoretical term, and one might think that attributing a property to someone 'under a visual mode of presentation' is enough to knock all worlds where that person lacks the property out of one's doxastic set.¹

The real problem is that (3) also seems true in the imagined scenario:

- (3) Joe believes that Eve is not involved with anyone.

Given (1), the only way these could both be true (in a single context) would be for $\text{DOX}(@, \text{Joe})$ to be the empty set, in which case all sentences of the form 'Joe believes P ' would be true.²

(From now on I'll go along with the assumption that in this case, the proposition **Eve is not involved with anyone** is true at all Joe's doxastic worlds and **Eve is involved with a linguist** is not. All that really matters is that at least one of the two propositions is not dox-ized.)

1.2 A standard descriptivist fix

The standard fix, inspired by Kaplan, is to somehow arrange the semantics so as get the following effect, where $\llbracket A \rrbracket$ is $\lambda w. \mathbf{a}$ for some $\mathbf{a} : e$.

- (4) $\llbracket \text{believes } A \text{ is a good dancer} \rrbracket(w, y) = 1$ iff for some individual concept $\mathbf{f} : s \rightarrow e$:

¹ Or one might have a contextualist spin on the Hintikka semantics, where different functions DOX are used in different contexts.

² One could try to reconcile this with Hintikka semantics by saying that (2) and (3) require different contexts. But this is hard to square with the truth of 'Ann and Bea both think that Eve is involved with a linguist' in a case where Ann only has a visual guise and Bea only has a name-and-past-acquaintance guise.

1. f is “appropriate for de re thought” by the standards of c .
2. $f(w) = \llbracket A \rrbracket(w)$
3. $\llbracket \text{is a good dancer} \rrbracket(f(v), v) = 1$ for all $v \in \text{Dox}(S, w)$.

This presentation fits most naturally with a semantics for definite descriptions like ‘the person dancing’ where they have denotations of type $s \rightarrow e$: then we can represent particular individual concepts by definite descriptions that denote them.

Choice points:

1. Is ‘believes that Eve is involved with a linguist’ *structurally ambiguous* between a reading with the plain Hintikkan truth-condition, and one with the ‘guise-quantification’ truth condition?³

Standard approaches say *yes*, since they generate the de re reading by positing optional meaning-shifting operations. But I know of no syntactic evidence for this!

M&S say no. I agree with them.⁴

2. Are there some contexts in which *all and only* constant individual constants are appropriate?

If so, then the (4)-style truth conditions relative to those contexts will coincide with the Hintikkan ones. (So, the claim that there is a possible reading for which the Hintikka truth conditions are correct does not settle the case in favour of the existence of the controversial structural ambiguity.)

3. Are constant individual concepts appropriate relative to all c ?

If so, all the possible *de re* truth conditions will be entailed by the Hintikkan one. For example: anyone who doxizes $\llbracket A \text{ is involved with a linguist} \rrbracket$ will satisfy ‘believes that A is involved with a linguist’, irrespective of the context.

I think M&S accept this implication.⁵

4. Does something analogous happen in types other than e ? (Surely yes! But let’s postpone this for now.)
5. For which intensional operators are the guise-quantification truth conditions available? The standard answer, I think, is that speech and attitude predicates, and maybe epistemic modals and indicative conditionals, but not circumstantial modals or counterfactual conditionals.⁶

M&S don’t discuss any such limitations, and their suggestion that their approach turns on ‘properties of meaning *per se*’ would make them hard to justify.

³ Note that this is *not* just the familiar scope ambiguity that arises when quantificational DPs occur under ‘believes’. The alleged ambiguity would be present even in sentences like ‘Someone at the party is such that Joe believes she is involved with a linguist’.

⁴ Though I have some doubts about their argument against structural ambiguity: see section 5 below for more.

⁵ See their discussion of example (37). I disagree, modulo the operative assumptions about doxizing. If everyone at the party saw Eve dancing and thought ‘That person is not single’, there’s a true reading of ‘No-one who saw Eve at the party thought she was single’, even if the guests include Joe who didn’t recognise her.

⁶ If no-one could have danced without blowing their cover, and Ann in fact the one person who danced but could have decided not to dance, ‘Ann couldn’t have avoided blowing her cover’ doesn’t seem to have a true reading (setting modal subordination aside).

2 Basics of Mayr and Schmitt’s approach

- For each expression A , the semantics keeps track of its “ordinary” semantic value $\llbracket A \rrbracket^o$ as well as a certain set $\llbracket A \rrbracket^t$ of “transparency

values” (of the same type). The latter is highly contextually variable, though subject to certain constraints which keep the transparency values ‘similar as regards contextually relevant questions’ to the ordinary values.

- The ordinary value of an expression is a member of its transparency value.
- Generally, the two tracks proceed independently, with pointwise composition for the transparency sets.
- *Exception:* the ordinary value of ‘S thinks P’ is the disjunction, for each proposition p in the transparency set of P , of the propositions that S doxizes q .⁷

Example: in the context of (2) the ordinary value of ‘Eve is involved with a linguist’ is the set of worlds where Eve is involved with a linguist. But its transparency value includes, inter alia, the set of all worlds where the person dancing is involved with a linguist. Since Joe doxizes the latter set, (2) gets to be true.

⁷ What’s the *transparency* value of this sentence? In the main text M&S make it the singleton of its ordinary value, though in footnote 17 they allow ‘think’ to itself have a nontrivial transparency set. It’s not clear to me that this is enough to handle the complexities of nested attitude reports.

3 *Bad intensions*

Our most fundamental concern about M&S’s approach is that it gives the wrong *intensions* to the relevant sentences. For example, in the posited context where (2) is true, (2) is predicted to express a proposition that is true at all worlds where Joe doxizes the proposition that the person dancing is involved with a linguist. (Including worlds where Eve isn’t even at the party!)

This seems problematic. One way to make the case is to embed the sentence under a further, circumstantial modal.

Scenario: Joe went to the party. It was certain that he’d see a person dancing with a linguist, and as a result, come to believe that the dancing person is involved with a linguist. But it was fifty-fifty whether the dancing person would be Eve or Ann. In fact, Eve was the dancing person, and Ann did not dance at the party.

(5) Joe was certain to believe that Eve was involved with a linguist.

(5) seems false here.

M&S could explain this by positing that in this new context, the transparency value of ‘Eve is involved with a linguist’ does not contain the proposition that the person dancing is involved with a linguist. But since there was no relevant uncertainty about the contents of Joe’s doxastic set, no posit about transparency values can explain why (6) seems to have a prominent true reading in the scenario:

(6) It was fifty-fifty whether Joe would believe that Eve was involved with a linguist.

Another way to bring out the worry is to use *propositional anaphora*. Suppose that after an earlier utterance of (2) (in the original context), someone follows up with

- (7) That's true, though it wouldn't have been true if Eve had stayed home tonight.

This seems like a good speech, even if things are set up such that if Eve had stayed home, Ann would have danced instead and Joe would still have doxized that the person dancing is involved with a linguist.

4 Constraining alternatives using QUDs

Scenario: Bob knows Eve, Ann and Bea well. If you asked him about Eve's relationship status, he would reply 'Eve is involved with Bea and not involved with Ann'. He is sitting by himself and not intuitively involved in any cases of recognition failure or identity confusion with respect to these people. We don't trust Bob's judgment, and have no idea whether Eve is involved with Ann, Bea, both, or neither.

- (8) Bob thinks that Eve is involved with Ann.

(8) does not seem to have any true readings in this context. But on M&S's approach, it would be true if the transparency value of 'Eve is involved with Ann' included the proposition that Eve is involved with Bea. So they need to posit a constraint that rules out that sort of transparency value.

I believe this is the idea.

QUD constraint For each $p \in \llbracket S \rrbracket^t$, one of the following conditions must obtain.⁸

1. (a) There is some question Q under discussion such that $p \cap c$ and $\llbracket S \rrbracket^o \cap c$ both entail the same complete answer of Q ('resolve Q in the same way'), and
 (b) There is no question Q under discussion such that $p \cap c$ and $\llbracket S \rrbracket^o \cap c$ entail incompatible partial answers to Q ('address Q differently').
2. [A different condition, which I won't go into since it will only make the view even more permissive.]

To explain why **Eve is involved with the person she is dancing with** satisfies condition (a) when S is 'Eve is involved with Ann', M&S say that both of these propositions address the question *Is Eve single?*, which is under discussion. This is presumably also true for **Eve is involved with Bea**.

Meanwhile, condition (b) will automatically be satisfied by any p such that $p \cap \llbracket S \rrbracket^o \cap c \neq \emptyset$ —including **Eve is involved with Bea**.⁹

Maybe instead of (b) we should have

⁸ What they actually say is just that there must be *some* $p \in \llbracket S \rrbracket^t$ that meets the given condition. But I think universal quantification must be intended, since the existential version would have no hope of preventing any undesired propositions from being in the transparency value.

⁹ Since Eve's being involved with both Ann and Bea is a live possibility.

(b') For every question Q under discussion and any complete answer q to Q , $p \cap c \cap q = \emptyset$ iff $\llbracket S \rrbracket^o \cap c \cap q = \emptyset$. (' p and $\llbracket S \rrbracket^o$ say the same thing about Q .')

Eve is involved with Bea will fail this condition, since it rules out the complete answer **Eve is involved with Ann and no-one else**, which **Eve is involved with Ann** does not rule out. **Eve is involved with the person she is dancing with** will meet (b') in a context where we *know* Ann is the person Eve is dancing with.¹⁰

In fact, no condition that just has to do with the test proposition p , context c , the questions under discussion, and the ordinary semantic value $\llbracket S \rrbracket^o$ will explain why 'Joe believes Eve is involved with Ann' is true in this scenario, but not in the scenario where our knowledge is the same but in fact Eve was dancing with Bea and not Ann. We need to look at the world of evaluation, as the standard descriptivist approach does.¹¹

¹⁰ Since then **Eve is involved with Ann** $\cap$ c = **Eve is involved with the person she is dancing with** $\cap$ c .

¹¹ This is another manifestation of the issue from section 3.

5 Narrow-scope transparent readings

A central contention of the paper is that the very same mechanism that can account for examples like (2) can also account for 'Narrow-scope Transparent' (NT) readings¹², as illustrated here:

Scenario: Joe believes that among the several people Eve was dancing with at the party there is someone she is involved with. In fact Eve was dancing with all and only the linguists. Joe has no idea that there were any linguists at the party.

(9) Joe thinks Eve is involved with a linguist.

M&S can say that **person Eve was dancing with** $\in \llbracket \text{linguist} \rrbracket^t$.

This will lead to similar objections to those we raised above for the treatment of 'de re' cases. But they also make certain interesting claims which are independent of that positive proposal:

1. The mechanism responsible for generating narrow scope transparent readings is the same one that allows sentences like (2) to be true.¹³
2. There is no *structural ambiguity* between the interesting (narrow-scope-transparent) and boring (narrow-scope-opaque) reading of (9).¹⁴

Claim (2) runs counter to most existing treatments of the NT phenomenon, which (in one way or other) turn on positing structural ambiguities.

M&S's argument against structural ambiguity turns on the following case.

¹² AKA 'Fodor's Third Readings'

¹³ One worry about this is that NT readings arise for circumstantial modals and counterfactual conditionals (e.g. 'It could have happened that every rich person was poor'), whereas these constructions do not seem to allow any analogue of the 'guise-quantification' phenomenon.

¹⁴ There's also the wide-scope transparent reading, 'There is a linguist that Joe believes Eve is involved with'. I don't read M&S as denying that this really does require some different resolution of structural ambiguity.

Scenario 4: Joe and Bob went to a party. Bob has no idea if there were any linguists at the party, but is convinced that Eve is in a relationship with a linguist. Joe thought all guests were biologists. Except for three linguists called Ann, Bea, and Cate, this was the case. Joe doesn't know them or their names. He saw each of them dancing with Eve, who he knows. So he thinks Eve and one of them are lovers but is not sure which of them. We are discussing if Eve is single; no other guest has an opinion.

They judge that in this case, sentences (10) are odd or unacceptable, whereas (11) is acceptable:

(10) (i) Exactly one guest thought that Eve is involved with a linguist.

(ii) Only one guest thought that Eve is involved with a linguist.

(iii) Only Joe thought that Eve is involved with a linguist.

(11) Two guests thought that Eve is involved with a linguist.

A proponent of structural ambiguity could agree that (11) has a true interpretation, if they think the NT disambiguation can be interpreted so that both Bob and Joe fall under it. And they could even agree that (iii) lacks a true interpretation, if they think the NT disambiguation can *only* be interpreted so that both Bob and Joe fall under it. But clearly, if there is a 'de dicto' (narrow-scope-opaque) disambiguation of 'thought that Eve is involved with a linguist', it applies to Bob and not Joe. So if (i) and (ii) really lack true readings, that is terrible news for the structural ambiguity thesis.¹⁵

To my ear, (i) and (ii) sound fine in this scenario.¹⁶ Meanwhile, (11) is at least a bit strange. I agree that (iii) is odd, though I suspect that this is due to some specifics of the case. In other examples one can easily hear true readings for analogues of (iii):

Scenario: we are playing a social deduction game. The evil players are Ann, Bea, and Cate; everyone else is good. Everyone knows there are three evil players including one demon. The good team are doing poorly: everyone except Joe is convinced that none of Ann, Bea, and Cate is the demon, while Joe thinks that two of them are good and the other is the demon (though even he has no view as to which is which).

(12) Only Joe thinks that some evil player is the demon.

So, I'm not yet convinced that there isn't some sort of structural ambiguity in play in generating NT readings.¹⁷

Is there any *positive argument* for such a structural ambiguity? It's tricky to pull apart from mere context-sensitivity. But one possible strategy is to look at judgments of *validity*: if an argument feels to have a valid reading, that suggests its premises and conclusion have a structural disambiguation that makes it valid.

I think we can find this with NT readings:

¹⁵ But I don't understand how M&S's own view is supposed to explain the supposed lack of true reading for (i) and (ii) in the scenario. Aren't we free to choose a small transparency set for 'Eve is involved with a linguist' that doesn't include any proposition doxized by Joe?

¹⁶ Imagine we heard Bob saying 'Eve is involved with a linguist' and Joe saying 'Eve is not involved with a linguist'.

¹⁷ Note that the standard ways of accounting this are set up in the context of Hintikka semantics, and will need modifications to allow for guise quantification for those who think that's a real phenomenon. 'There is a linguist Joe thinks Eve is involved with' plausibly entails the NT-reading of 'Joe thinks Eve is involved with a linguist', although it could be true if Joe only doxizes that Eve is involved with the person she is dancing with (and has belief worlds she is not involved with any actual linguist).

- (13) a. Joe thinks that Eve is involved with a linguist.
b. All and only the linguists are fans of K-pop.
c. So, Joe thinks that Eve is involved with a fan of K-pop.¹⁸

¹⁸ Schema: S Att $\phi[F]$; all and only F s are G s; therefore S Att $\phi[G]$.